

The Photofragmentation of Simple van der Waals Complexes

Classical Statistical Theory and Comparison with Exact Classical and Approximate Quantum Dynamics Results

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A recently proposed statistical theory for unimolecular reactions, based on employing approximate phase-space bottlenecks as transition states, is discussed and a new and simpler expression for the rate constant is derived. The theory is applied to a collinear model of van der Waals molecule fragmentation and the results are compared with exact classical and approximate quantum-mechanical rate constants. Sources of non-statistical behaviour and classical/quantum-mechanical discrepancy are discussed.

The photofragmentation of van der Waals complexes is usually cited as one of the few examples of a non-statistical unimolecular reaction.¹ However, a recently proposed alternative version of the standard statistical theory of unimolecular reactions (RRKM theory) can in fact describe some aspects of the fragmentation of HeI₂ van der Waals complexes:²



where v denotes the effective vibrational quantum number in the I₂ part of the complex and v' is the product vibrational quantum number. The new theory was motivated by the work of Davis and Gray,³ in which the notion of an intermolecular bottleneck in the molecular phase space was introduced *via* analysis of a T-shaped model of HeI₂ fragmentation. The exact intermolecular bottleneck rigorously surrounds the region in phase space where unimolecular complexes, as opposed to approaching and departing fragments, exist. Furthermore, the exact intermolecular bottleneck is a perfect transition state⁴ in the sense that any phase point that crosses it in the outward direction never returns, and therefore forms products. In the alternative RRKM theory an approximation to this bottleneck is used to define the transition-state surface and the fragmentation rate constant is given by the ratio of the outward flux of classical phase points through this surface to the concentration of classical phase points within the surface.

It should be emphasized, of course, that van der Waals molecules do not, in general, fragment as described by any simple statistical theory and that a truly quantitative understanding of the process requires rigorous quantum-dynamics calculations.⁵ Nonetheless, a careful study of when and why approximate statistical theories, such as the alternative form for RRKM theory, do describe some aspects of the dynamics is very useful for obtaining a physical picture of the fragmentation process. Such a study involves investigating the validity of the three basic assumptions made by the alternative RRKM theory, specifically: (i) the adequacy of classical mechanics to describe the molecular motions; (ii) the accuracy of the approximation to the intermolecular bottleneck as a transition state; and (iii) the existence of rapid intramolecular energy exchange, compared to the rate of the unimolecular reaction under consideration, for any initially prepared ensemble of phase points within the approximate intermolecular bottleneck.

The purpose of the present paper is twofold. First, we derive an approximate form for the alternative RRKM rate constant that shows clearly the qualitative trends in the

fragmentation rate as a function of intramolecular potential-energy parameters. Secondly, we examine critically the above three assumptions for a collinear model of ArI_2 fragmentation. The ArI_2 system, because of the large noble-gas atom mass and relatively large van der Waals well depth, is expected to provide a better candidate for making classical/quantum correspondence arguments than, say, HeI_2 . Furthermore, the dynamical coupling in the ArI_2 model we study is larger than in HeI_2 and leads to interesting observations related to assumptions (ii) and (iii).

In the next section we discuss the alternative RRKM theory and present a derivation of a new approximate form for the rate constant that is simpler and allows examination of the qualitative behaviour of the fragmentation rate as a function of intramolecular potential-energy parameters. We then apply the theory to obtain the rate of fragmentation of a collinear model for ArI_2 and compare the results with fragmentation rates computed from classical trajectories and also with rates computed from an approximate quantum-mechanical Fermi golden rule approach. Finally we summarize our analysis and suggest some future applications of the methods discussed in this paper.

The Alternative RRKM Theory

General Formulation

We consider a molecular system described by a classical Hamiltonian H . The classical transition-state-theory rate constant for a unimolecular reaction at an energy E is^{2,6}

$$k_s(E) = \frac{F_s(E)}{N_R(E)} \quad (2)$$

where F_s is the flux of phase points in the direction of products that cross a transition state dividing surface S in the phase space and N_R is the relevant measure of reactant phase-point concentration. Thus

$$F_s(E) = \int d\mathbf{p} \, d\mathbf{x} \, \delta(S) \delta(E - H) \dot{S} \theta(\dot{S}) \quad (3)$$

and

$$N_R(E) = \int d\mathbf{p} \, d\mathbf{x} \, \delta(E - H) \theta(-S). \quad (4)$$

Note that through the step function θ , defined such that $\theta(z > 0) = 1$ and $\theta(z < 0) = 0$, the reactant phase space in eqn (4) is defined to be all the phase space within the transition-state surface

$$S(\mathbf{p}, \mathbf{x}) = 0 \quad (5)$$

where the convention $S < 0$ implies a phase point $(\mathbf{p}, \mathbf{x})$ is within the surface and $S > 0$ implies the point is outside the surface. As noted in ref. (2) and (6), one need not restrict the definition of the reactant phase space to be the entire phase space within the transition-state dividing surface S ; however, this assumption is consistent with both the alternative RRKM theory and most applications of standard RRKM theory. The time derivative in eqn (3) may be obtained from the Poisson bracket relation⁷

$$\dot{S} = \sum_k \left(\frac{\partial S}{\partial x_k} \frac{\partial H}{\partial p_k} - \frac{\partial S}{\partial p_k} \frac{\partial H}{\partial x_k} \right). \quad (6)$$

Standard RRKM theory results^{2,6} if one assumes $S = R - R^\ddagger$, where R is the relevant reaction coordinate and $R^\ddagger$ is a particular (transition-state) value of the reaction coordinate such that products are assumed to form directly if $R > R^\ddagger$. Implicit in this choice of critical surface is that unimolecular complexes exist for all the classical phase space consistent with energy conservation and the condition $R < R^\ddagger$.

It is important to understand clearly the assumptions inherent in eqn (2). The flux F_s represents the instantaneous rate of flow of phase points out of the transition-state surface. It would only correspond to the true reactive flux if none of the phase points that exit the transition state surface come back. This is of course the well established⁴ 'point of no return' property that the best transition-state surface must satisfy. The fact that one assumes a microcanonical distribution within the transition-state functions, as represented by the energy delta functions in eqn (2) and (3), is consistent with the notion of equilibrium (*i.e.* rapid intramolecular energy exchange) of any initially prepared ensemble on a timescale shorter than that required for the unimolecular reaction under consideration. The region of phase space within the transition-state surface can therefore be associated with the region where unimolecular complexes exist.

The standard RRKM theory assumption that unimolecular complexes exist for all the phase space such that $R < R^*$ is a good one when the molecule can be represented as a set of strongly coupled oscillators. However, as shown in ref. (2) and (3), this assumption is not valid for van der Waals molecules. The van der Waals bond energy is much smaller than typical molecular bond energies, a fact which, along with the weak coupling, leads to only a small portion of the phase space being occupied by unimolecular complexes. A more reasonable criterion for the transition-state surface that surrounds the phase-space region where complexes exist is

$$S(P, R) = h_R(P, R) = 0 \quad (7)$$

where

$$h_R(P, R) = \frac{P^2}{2\mu} + V_R(R). \quad (8)$$

P and R are the canonical momentum and position variables associated with the van der Waals degree of freedom, μ is the effective mass and V_R is the effective interaction potential. Eqn (7) is the condition that the effective energy in the reactive degree of freedom be equal to zero, where the energy units are chosen such that zero energy in h_R corresponds to there being exactly the required dissociation energy in the reactive degree of freedom. Eqn (7) may thus be thought of as a minimum energy requirement for dissociation. It is interesting to note that some of Wigner's⁸ earliest transition-state theory studies also employed similar ideas. A more interesting interpretation of eqn (7), however, is that it defines a phase-space trajectory known as a separatrix.⁹ Fig. 1(a) depicts the classical trajectories associated with the zero-order hamiltonian, eqn (8). Each curve is a solution of $h_R(P, R) = \epsilon$. The separatrix, $\epsilon = 0$, is the dashed curve which separates the purely bound (closed) from unbound or scattering (open on the right side) trajectories.

In a system with coupling between the degrees of freedom, much of the structure in fig. 1(a) is destroyed and many phase points that were constrained to be bound forever in the uncoupled system now may escape. However, the remnants of the zero-order separatrix act as a partial barrier to the flow of phase points out of the interaction region. Fig. 1(b) illustrates this effect by displaying a composite surface of section obtained by propagating twenty trajectories in the ArI_2 model system that is discussed in detail in the next section. It can be seen that the phase points do roughly fill out the interior of the zero-order separatrix. All the trajectories shown, however, do eventually escape from the interior of the zero-order separatrix and form products. In fact, while it is not evident from the static surface of section representation, phase points exit the separatrix region *via* the first finger-like structure on the left and then evolve from finger to finger, from left to right, never returning to the interaction region.

Actually, the separatrix shown in fig. 1 is not an exact transition state, although it is certainly very close to such a structure. Davis and Gray,³ for problems with two degrees of freedom, were able to provide a procedure for constructing an exact transition

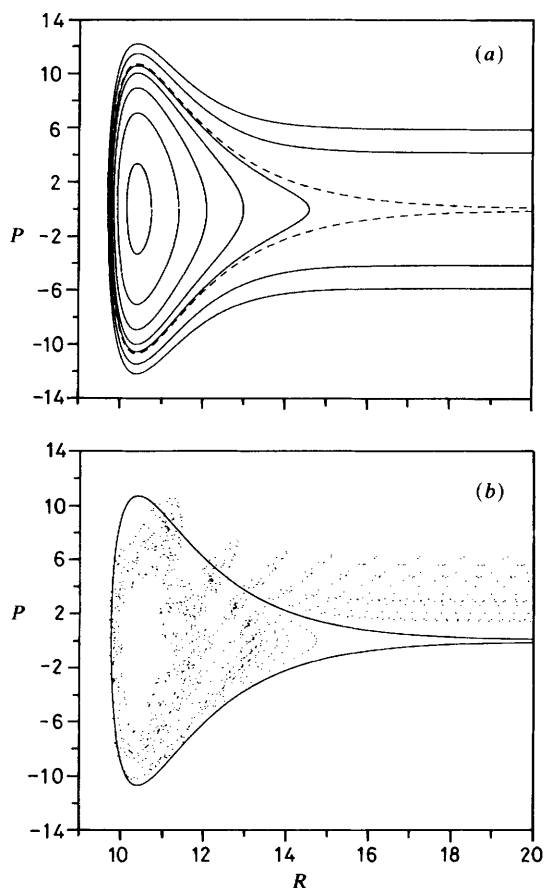


Fig. 1. (a) Classical phase space curves for the zero-order hamiltonian, h_R . The zero-order separatrix is indicated by a dashed curve. The parameters for h_R are those for ArI_2 given in the text. (b) Surface of section for twenty trajectories for ArI_2 . P and R are plotted each time $r = \bar{r}$ and $p > 0$. The zero-order separatrix is drawn in as a solid curve. The energy for all the trajectories is $E = 107.3 \text{ cm}^{-1}$, corresponding to $v = 1$, $n = 3$.

state, *i.e.* an exact surface of no return. The exact transition state is the analogue of a separatrix when dynamical coupling exists and will also be referred to as the exact separatrix (as opposed to the zero-order separatrix discussed above). The method used to construct the exact separatrix is taken from non-linear dynamics,^{9–12} in particular the work of Channon and Lebowitz.¹² To supplement the detailed exposition of Davis and Gray,³ we give a succinct prescription for obtaining the exact separatrix, employing a little more explicitly the mathematical language of non-linear dynamics. The essential idea is that when non-trivial coupling exists in a dynamical system, the zero-order separatrix may be destroyed in the sense that it is no longer an invariant manifold in the phase space. The destruction of the zero-order separatrix leads to a kind of chaos which, in the present situation, corresponds to unimolecular complexes that ultimately fragment. The exact separatrix for problems with two degrees of freedom can be constructed on a surface of section and is precisely the union of portions of the global stable and unstable invariant manifolds¹¹ of a relevant fixed point at a homoclinic point (a point of intersection of the manifolds). Fig. 2 illustrates the construction of the exact separatrix for a relevant energy of the ArI_2 problem discussed in the next section. The

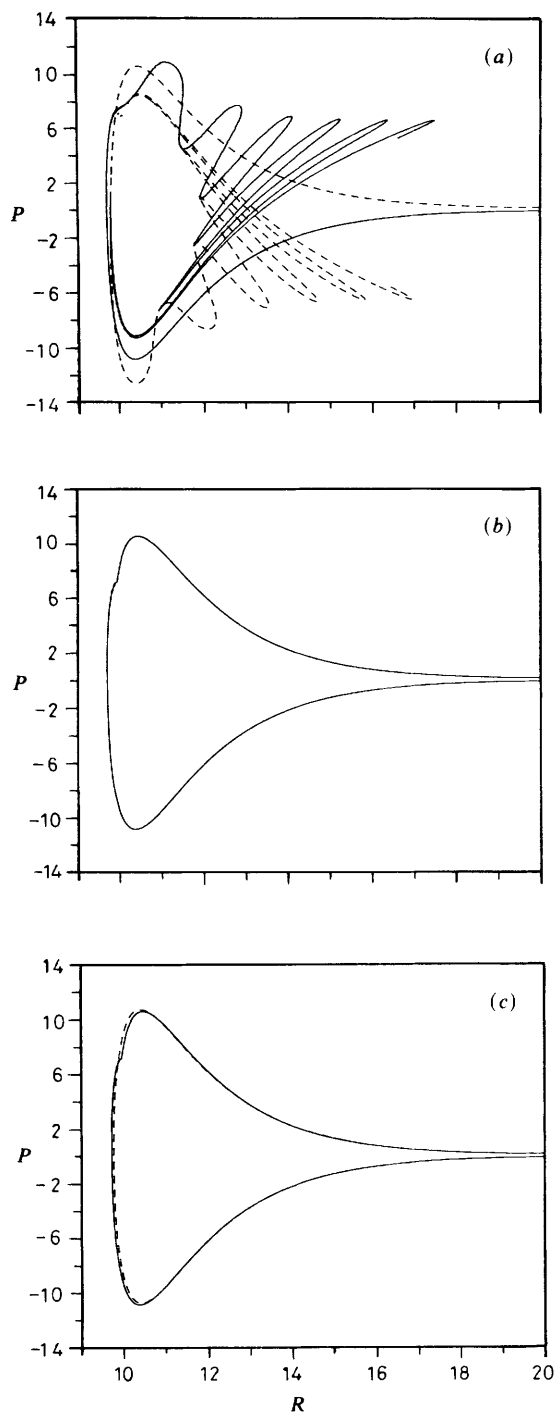


Fig. 2. Construction of an exact separatrix or transition state. (a) The stable (---) and unstable (—) manifolds for $v=1, n=3$ for ArI_2 . (b) A particular union of the manifolds, suitable to be taken as the exact separatrix. (c) Comparison of the exact separatrix from (b), and the zero-order separatrix of fig. 1, drawn as a dashed curve.

stable (dashed curve) and unstable (solid curve) manifolds are displayed. The fixed point in this case exists at $R = \infty$, $P = 0$. Note that in the limit of large R the stable and unstable manifolds are essentially the same as the positive and negative momentum zero-order separatrix branches. The construction of each of these manifolds involves the propagation in time of *ca.* 100 classical trajectories and is straightforward for two degree of freedom problems; the explicit procedure for their construction is described by Davis and Gray.³ If there were no coupling, the stable and unstable manifolds would join one another smoothly and form a zero-order separatrix. In the presence of coupling the two manifolds intersect in increasingly convoluted ways, each intersection being called a homoclinic point. The mess of intersections shown in fig. 2(a) is called the homoclinic tangle¹¹ and is the source of chaotic behaviour. The union of two portions of the stable and unstable manifolds shown in fig. 2(b) surrounds all the homoclinic tangle. Furthermore, Davis and Gray³ show that because the manifolds are invariant and phase space areas are conserved, the structure in fig. 2(b) has the desired transition-state properties. Thus if a phase point exits the interior of the exact separatrix in fig. 2(b), it does not return. There is a certain ambiguity in the definition of the exact separatrix in that alternative and equivalent separatrices, based on other homoclinic points, can be constructed. Another important aspect of this construction is that it is easy to estimate the flux out of the exact separatrix numerically. One merely propagates the entire structure in fig. 2(b) one surface of section time unit. The area that escapes from this structure is, in fact, the flux F_s . A useful property of the exact separatrix is that the escaping area is localized in a small region near where the stable and unstable manifolds were joined; this area may be thought of as having passed through a turnstile.³

At this stage the reader may wonder why one bothers at all with a theory based on a zero-order separatrix. There are two reasons for generating such a theory. First, it is the simplest approach because it involves no dynamics; the integrals of eqn (3) and (4) determine the rate constant. Secondly, extensions to systems with more than two degrees of freedom are feasible with a zero-order separatrix analysis. At present, it is not obvious how to set up a numerically tractable procedure for obtaining exact separatrices for problems with three or more degrees of freedom. Fig. 2(c) compares the exact (solid) and zero-order (dashed) separatrices for a particular energy in the ArI₂ system. While the differences are seen to be small, we will nonetheless point out later how the use of the zero-order separatrix can lead to errors in calculating the flux.

As noted above, the alternative version of RRKM theory can be applied to systems with several degrees of freedom. One can always, for example, analytically evaluate portions of the integrals in eqn (3) and (4) by making use of the rules of delta-function integration; the remaining integrals may be evaluated *via* Monte Carlo procedures. The required input is a reasonable representation of the potential energy V and a good estimate of the effective potential in the reactive degree of freedom V_R . In this respect our theory differs from simple versions of RRKM theory¹³ that do not include the full potential energy explicitly in the rate expression and only invoke its existence in order to argue for rapid redistribution of phase points.

In the situation where one has just one molecular degree of freedom, r , in addition to the reactive degree of freedom, R , with a full hamiltonian of the form

$$H = \frac{P^2}{2\mu} + \frac{p^2}{2m} + V(R, r) \quad (9)$$

eqn (2), along with the zero-order separatrix eqn (7), reduces to²

$$k_s(E) = \frac{(2m)^{1/2}}{N_R(E)} \int \frac{|\partial(V - V_R)/\partial R|}{[E - (V - V_R)]^{1/2}} dR dr. \quad (10)$$

The integration range above is all R and r such that $V_R(R) \leq 0$ and $V(R, r) - V_R \leq E$. The two-dimensional integral in eqn (10) may be evaluated *via* numerical integration;

this was the approach taken in ref. (2). If the energy in the reactive degree of freedom can be considered small in relation to the energy in the molecular degree of freedom, it is easy to see that eqn (4) for N_R can be approximated by²

$$N_R(E) = A_s \frac{2\pi}{\omega_r(E)} \quad (11)$$

where A_s is the phase space area (or action) within the zero-order separatrix and $\omega_r(E)$ is the classical vibrational frequency in the non-reactive degree of freedom if all the energy were in it. For the van der Waals problems of interest to us here, eqn (11) is accurate to within a few percent.

A Simple Approximation for the Rate

We now show that a simpler form for the alternative RRKM theory rate results under certain approximations. For simplicity we consider again a system with two relevant coordinates. Eqn (9) may be expressed as

$$H = H_0(P, R, p, r) + W(R, r) \quad (12)$$

where the zero-order hamiltonian is

$$H_0 = h_R(P, R) + h_r(p, r), \quad (13)$$

with h_R given by eqn (8) and

$$h_r = \frac{p^2}{2m} + V_r(r). \quad (14)$$

The coupling is given by

$$W(R, r) = V(R, r) - V_r(r) - V_R(R). \quad (15)$$

If the momentum integrals in the flux integral eqn (3) are evaluated explicitly using the rules of delta-function integration, eqn (10) results. However, instead of doing this we now perform a change of variables from (p, r) to the action-angle variables^{7,14} (v, q_v) . As usual, v is the classical analogue of a quantum number and is proportional to an action integral, while the angle q_v is a measure of the vibrational phase. The transformation to action-angle variables reduces the zero-order Hamiltonian in the non-reactive degree of freedom to just a function of the action v : $h_r[p, (v, q_v), r(v, q_v)] = \varepsilon_r(v)$. Eqn (3) then becomes

$$F_s = \int_0^\infty dP \int_{R_s}^\infty dR \int_{-1/2}^{v_{\max}} dv \int_0^{2\pi} \delta(S) \delta(E - H) \frac{P}{\mu} \left| \partial W / \partial R \right| dq_v \quad (16)$$

where the integration range is such that R goes from the inner turning point of the separatrix, R_s , to ∞ . R_s is the smallest solution of $V_R(R_s) = 0$. The absolute value term arises from evaluating S and noting the relation

$$\int_{-\infty}^\infty \theta(Pf) P f g(P) dP = \int_0^\infty P |f| g(P) dP$$

where g is an even function of P and f is a function of R and r that may take on either positive or negative values. Now

$$H = h_R(P, R) + \varepsilon_r(v) + W(R, v, q_v) \quad (17)$$

and assuming that the coupling energy W is small enough to be neglected in comparison with E and ε_r , keeping in mind that $\delta(S)$ selects out values of (P, R) such that $h_R = 0$, the function $\delta(E - H)$ may be approximated by

$$\delta(E - H) \approx \frac{\delta(v - v_E)}{\omega_r(E)} \quad (18)$$

where v_E is the vibrational action if all the energy were in the non-reactive degree of freedom and $\omega_r(E)$, the corresponding classical frequency of vibration in the non-reactive degree of freedom, is related to ε_r through the standard relationship^{7,14} $\omega_r = \partial \varepsilon_r / \partial v$. Insertion of eqn (18) into eqn (16) gives

$$F_s = \frac{1}{\omega_r(E)} \int_0^\infty dP \int_{R_s}^\infty dR \int_0^{2\pi} \delta(S) \frac{P}{\mu} \left| \partial W(R, v_E, q_v) / \partial R \right| dq_v. \quad (19)$$

We now use the fact that

$$\delta(S) = \sum_k \frac{\mu}{|P_k|} \delta(P - P_k) \quad (20)$$

where $P_k = \pm(-2\mu V_R)^{1/2}$, to obtain

$$F_s = \frac{1}{\omega_r(E)} \int_0^{2\pi} dq_v \int_{R_s}^\infty \left| \partial W(R, v_E, q_v) / \partial R \right| dR. \quad (21)$$

We expect that the main contribution to the integral over R is in a region where $\partial W / \partial R$ does not change sign. We have verified, by explicit calculation for the van der Waals systems studied in this paper, that this assumption is indeed justified. Then, since in the limit of infinite separation of the fragments $W = 0$

$$\int_{R_s}^\infty \left| \partial W / \partial R \right| dR \approx |W(R_s, v_E, q_v)| \quad (22)$$

so that eqn (21) becomes

$$F_s(E) = \frac{1}{\omega_r(E)} \int_0^{2\pi} |W(R_s, v_E, q_v)| dq_v. \quad (23)$$

Finally, eqn (23) along with eqn (11) for N_R leads to

$$k_s(E) = \frac{1}{2\pi A_s} \int_0^{2\pi} |W(R_s, v_E, q_v)| dq_v. \quad (24)$$

Eqn (24) states that the rate constant is proportional to a phase average over the non-reactive degree of freedom of the magnitude of the coupling, with the reactive degree of freedom held fixed at its (separatrix) inner turning point value. Eqn (24) is a simple one-dimensional integral for the rate constant and, although more approximate than eqn (10), is easier to evaluate and leads to further analytical simplifications. It is also similar in spirit to certain quantum-mechanical Franck-Condon or reflection¹⁵ type approximations of the Fermi golden rule formulation of the reaction rate constant.

Table 1 displays the results of applying eqn (24) to the fragmentation of a T-shaped HeI_2 model discussed in our earlier work^{2,3,14} and contrasts its predictions with those obtained from eqn (10). It is clear that quite reasonable agreement is found. Ref. (2) contains a more detailed discussion of the alternative RRKM theory predictions for this system, and compares the results with dynamical rate constants calculated from classical trajectory studies. With the exception of the lowest energy in table 1, the alternative RRKM theory rate constants are within an order of magnitude of the dynamical results. Standard RRKM theory, as noted in ref. (2), cannot be unambiguously applied to such problems: the rate constant depends sensitively on the choice of the transition-state position $R^\ddagger$, and reasonable choices can lead to very bad results.

Table 1. Comparison of the approximate alternative RRKM rate constants [eqn (24)] with the more accurate alternative RRKM rate constants [eqn (10)] for the T-shaped HeI₂ fragmentation model

E/cm^{-1} ^a	v ^b	$\hbar k/\text{cm}^{-1}$ ^c	
		eqn (24)	eqn (10)
−3683	10	0.43	0.52
−2662	20	0.66	0.80
−1807	30	0.90	1.09
−574	50	1.43	1.89

^a Energy, in units such that the separated atoms have zero energy. ^b The effective quantum number in the I₂ part of the van der Waals complex, assuming zero-point energy in the van der Waals bond. ^c All rate constants are expressed as full widths at half maximum, i.e. $\Gamma = \hbar k$.

Qualitative Trends in the Statistical Rate Constant

Eqn (24) can be analysed in an approximate fashion to show certain trends in the rate constant with respect to variation of intramolecular potential-energy parameters. This discussion is a classical analogue of more sophisticated quantum-mechanical arguments of Beswick and Jortner.¹⁶ For the sake of concreteness, consider a collinear molecule X-BC, where X represents the noble-gas atom and BC is a non-rotating diatomic molecule. If R is the distance between X and the centre of mass of BC and r is the BC interatomic distance, then a reasonable Hamiltonian model is that studied by Beswick and Jortner:¹

$$H = \frac{P^2}{2\mu} + \frac{p^2}{2m} + V_{\text{XB}}[R_{\text{XB}}(R, r)] + \frac{1}{2}m\omega_{\text{BC}}^2(r - \bar{r})^2 \quad (25)$$

where R_{XB} is the distance between X and B, which is related to the chosen canonical coordinates through

$$R_{\text{XB}} = R - \frac{r}{2}. \quad (26)$$

Non-trivial coupling arises because V_{XB} , through eqn (26), mixes the coordinates R and r . The natural choice for the effective van der Waals interaction is then

$$V_{\text{R}}(R) = V_{\text{XB}}[R_{\text{XB}}(R, \bar{r})]. \quad (27)$$

V_{XB} is taken to be a Morse potential:

$$V_{\text{XB}}(R_{\text{XB}}) = D_{\text{XB}}\{\exp[-2\alpha_{\text{XB}}(R_{\text{XB}} - \bar{R}_{\text{XB}})] - 2\exp[-\alpha_{\text{XB}}(R_{\text{XB}} - \bar{R}_{\text{XB}})]\}. \quad (28)$$

The coupling W is approximated by expanding the potentials to quadratic order:

$$W(R_{\text{s}}, r) \approx D_{\text{XB}}\alpha_{\text{XB}}^2[(R_{\text{s}} - r/2 - \bar{R}_{\text{XB}})^2 - (R_{\text{s}} - \bar{r}/2 - \bar{R}_{\text{XB}})^2]. \quad (29)$$

Now r must be expressed in terms of its action-angle variables, with the action such that all the system energy E is placed in the diatom. Since V_{r} is harmonic, $E = (v_{\text{E}} + \frac{1}{2})\hbar\omega_{\text{BC}}$ and⁷

$$r = \bar{r} + (2E/m\omega_{\text{BC}}^2)^{1/2} \sin q_{\text{v}}. \quad (30)$$

Insertion of eqn (30) into eqn (24) leads to

$$k_s(E) = \frac{D_{XB}\alpha_{XB}^2}{2\pi A_s} \int_0^{2\pi} |\delta(2E/m\omega_{BC}^2)^{1/2} \sin q_v + \frac{1}{4}(2E/m\omega_{BC}^2) \sin^2 q_v| dq_v \quad (31)$$

where

$$\delta = \bar{R}_{XB} + \bar{r}/2 - R_s.$$

For many energies of interest and for van der Waals parameters such as those in the model for ArI₂ discussed in the following section, the first term in eqn (31) is an order of magnitude greater than the second. Neglecting the second term leads to

$$k_s(E) \approx \frac{\ln(2)\omega_{XB}\alpha_{XB}}{2\pi^2} (2E/m\omega_{BC}^2)^{1/2} \quad (32)$$

where we have explicitly evaluated the area of the separatrix for the Morse oscillator,² $A_s = 4\pi D_{XB}/\omega_{XB}$, and the parameter δ has been expressed in the equivalent form $\ln(2)/\alpha_{XB}$.

Eqn (32) is the desired result. It shows a number of interesting trends that were also noted by Beswick and Jortner¹⁶ in the approximate quantum-mechanical treatment of the same problem. First, one notes that as ω_{BC} decreases, the rate constant increases, which is one way of stating the energy-gap law.¹ The rate constant is also seen to increase with increasing curvature parameter α_{XB} , which is a measure of the steepness of the interaction potential. A final trend, also found by Beswick and Jortner, is that the rate constant will decrease as the van der Waals well depth D_{XB} is increased, keeping its frequency ω_{XB} fixed. [This can be seen by using the form $\alpha_{XB} = \omega_{XB}(\mu/2D)^{1/2}$.] Of course, the functional dependence inferred from this classical treatment is not the same as the quantum-mechanical result.¹ However, it is very interesting that many of the trends found in the behaviour of $k_s(E)$ are similar.

Application to Collinear van der Waals Molecule Fragmentation

Model and Methods

We examine in detail a collinear model for ArI₂ fragmentation discussed previously by Beswick and Jortner^{1,15} and Halberstadt and Beswick.¹⁷ The I₂ potential is taken to be harmonic and the ArI interaction is taken to be a Morse function so that the relevant full Hamiltonian is given by eqn (25). The potential parameters are: $D_{XB} = 200 \text{ cm}^{-1}$, $\alpha_{XB} = 2.02 \text{ \AA}^{-1}$, $\bar{R}_{XB} = 4 \text{ \AA}$, $\hbar\omega_{BC} = 128 \text{ cm}^{-1}$, $\bar{r} = 3.016 \text{ \AA}$, with masses $\mu = 34.52 \text{ amu}$ and $m = 63.5 \text{ amu}$. It should be emphasized that this representation is not an adequate model of real ArI₂ fragmentation. In addition to inaccuracies in the potential model and the restriction to collinear motion, there is also experimental evidence¹⁸ that electronic predissociation may compete with vibrational predissociation.

The alternative RRKM theory results were obtained by direct numerical integration of eqn (10), with eqn (11) used to obtain N_R . Use of the more approximate equation (24), discussed earlier, leads to results that are at most 15% smaller.

Classical dynamical rate constants are obtained in the usual fashion^{3,14} by considering ensembles of 200 phase points (P, R, p, r) for each energy E and numerically integrating Hamilton's equations of motion to obtain the survival probability, P_s , or probability of having not fragmented to Ar + I₂, as a function of time t . Fragmentation is said to have occurred if $R > 25a_0$ and $P > 0$. Linear fits to plots of $\ln(P_s)$ vs. t then yield approximations to the classical decay rates. In some cases there can be a time delay before P_s is less than one. Also, non-linearities may occur when long times are examined. Consequently, the fits to P_s are performed between the minimum time that fragmentation starts to occur and the time for which half the population has been depleted, i.e. when

$P_s = 0.5$. There are many ways of choosing initial ensembles consistent with a given total energy. We choose to use ensembles consistent with various initial excitations in the complex. The energy in the van der Waals degree of freedom, h_R , when expressed in terms of its action-angle variables, is:¹⁴

$$\varepsilon_R(n) = (n + \frac{1}{2})\hbar\omega_{XB} - (n + \frac{1}{2})^2\hbar^2\omega_{XB}^2/D_{XB} - D_{XB}. \quad (33)$$

Our samples are defined by specifying initial excitations corresponding to vibrational actions v in I_2 and n in the van der Waals degree of freedom. The total energy is taken to be

$$E = \varepsilon_R(n) + \hbar\omega_{BC}(v + \frac{1}{2}). \quad (34)$$

The vibrational phase angle in the van der Waals degree of freedom q_n is then varied between 0 and 2π in 200 even steps and for each step an initial condition is determined as follows: (P, R) are determined from the well known relationships¹⁴ $P(n, q_n)$, $R(n, q_n)$ for a Morse oscillator; r is set equal to its equilibrium value $\bar{r}$ and p is determined by energy conservation. With the present Hamiltonian and choices of zero-order Hamiltonians, this procedure yields exactly the desired energies initially in the two degrees of freedom. As time proceeds, of course, energy will be exchanged *via* the coupling W . The closed phase-space curves in fig. 1(a) represent certain of the ensembles studied. The (P, R) points corresponding to the $n = 0$ ensemble all lie on the innermost closed curve. It should be realized, however, that while the various ensembles correspond to a uniform distribution in vibrational phase q_n in the van der Waals degree of freedom, this corresponds to a distribution of points along the closed (P, R) curves with the maximum and minimum values of R , or inner and outer turning points, weighted more than others. The remaining closed curves in fig. 1(a) correspond, as one goes outwards from $n = 0$, to $n = 2, 6$ and 8.

From the point of view of quantum mechanics, van der Waals complexes are scattering resonances,^{1,16,18} whose full widths at half maximum are related to decay rates. Assuming W is small, the resonance positions are approximately determined by the eigenvalues of the zero-order Hamiltonian, eqn (13). Eqn (34) with n and v restricted to positive integers or zero is also the correct quantum-mechanical expression for the zero-order resonance positions. The parameters of the present model are such that n can take on only the values $n = 0, 1 \dots 9$. Since $\varepsilon_R(n+1) - \varepsilon_R(n) < \hbar\omega_{BC}$, it is useful to think of each v state of the complex as possessing a manifold of 10 van der Waals states. Quantum-mechanical decay rates can be approximated by the Fermi golden rule^{1,18} form:

$$k_{vn} = \frac{2\pi}{\hbar} \sum_{v'} |\langle vn | W | v' \varepsilon' \rangle|^2 \quad (35)$$

where $|v', \varepsilon'\rangle$ denotes a bound-continuum state of eqn (13), v' is a product I_2 Morse quantum state and $|\varepsilon'\rangle$ is an energy-normalized translational state of the van der Waals degree of freedom. The sum is over all allowed product states. Eqn (35) is evaluated using the analytical formulae for bound-bound and bound-continuum matrix elements already given by Beswick and Jortner.¹ Since the states that enter into eqn (35) do account partially for the distortion of the wavefunctions due to the van der Waals interaction, Beswick and Jortner refer to predictions from this equation as distorted wave results. There are some limited, but presumably accurate, quantum-decay rates available for the ArI_2 system, calculated by Beswick and Jortner^{1,16} and reported by Halberstadt and Beswick,¹⁷ obtained from coupled-channel scattering resonance widths. Table 2 compares these results with the more approximate results from eqn (35). Eqn (34) slightly overestimates the decay rate constants, by a factor ranging between 1.3 to 1.5 and relative trends are maintained well. The Fermi golden rule rate constants in table 2 are in essentially exact accord with the distorted-wave diabatic (DWD) rate

Table 2. Comparison of exact quantum (EQ) decay rate constants with the Fermi golden rule (FGR) estimates of eqn (35) for the collinear ArI₂ model

E/cm^{-1} ^a	v	n	$\hbar k/\text{cm}^{-1}$	
			EQ ^b	FGR
79.4	1	2	0.26	0.34
107.3	1	3	0.34	0.44
131.3	1	4	0.38	0.50
151.3	1	5	0.38	0.50
167.3	1	6	0.34	0.45
179.4	1	7	0.24	0.36
187.4	1	8	^c	0.23
191.5	1	9	^c	0.08

^a Energy zero chosen with Ar infinitely separated from I₂. ^b From ref. (17). ^c Result not given in ref. (17).

constants given by Halberstadt and Beswick,¹⁷ which is a check on our implementation of eqn (35).

Results and Discussion

Before examining the non-statistical effects that arise in the ArI₂ model system we wish to discuss a more basic approximation of the statistical theory, namely its approximation of the exact intermolecular separatrix as a zero-order separatrix. As noted earlier, the exact separatrix is a perfect transition state and its construction requires a substantial dynamical input.¹⁹ Fig. 2(c) compares the zero-order separatrix to the exact separatrix for a relevant energy in the ArI₂ system. While the deviations between exact and approximate separatrices appear small, the flux out of the separatrix itself is small compared to the area the separatrix encompasses, so that errors can arise. This is particularly so when the dynamical coupling is large. The flux across the exact separatrix is obtained by propagating the separatrix one unit of time on the surface of section, which corresponds to one period of the I₂ molecule, and then calculating the area that escapes the original closed curve.³ Note that this flux calculation involves the dynamical propagation of phase points. The statistical theory flux is obtained from eqn (3) as the instantaneous flux out of the zero-order separatrix and thus involves two inherent approximations. Note in particular that the statistical theory flux is not the same as a dynamical flux calculated from the zero-order separatrix by propagating it one unit of time on the surface of section and estimating the escaped area. Table 3 illustrates these points by comparing the fluxes obtained by propagation of the exact and zero-order separatrices with the statistical theory flux. The difference between the statistical theory flux and the exact separatrix flux is a little over a factor of two. In the HeI₂ system discussed previously² the difference was much less, owing presumably to the smaller dynamical coupling in the HeI₂ system. Another source of discrepancy that we will treat more carefully in future work is the difference between the phase-averaged flux obtained from the statistical theory [eqn (23)] and the flux obtained from the exact separatrix theory, which is evaluated on a surface of section which corresponds to a particular phase.

The approximate quantum-mechanical resonance decay widths are displayed in fig. 3 as squares connected by solid lines. Each inverted parabola pattern arises from a separate v manifold, with n ranging from 2 to 9. Quantum states with $n=0$ and 1 did not decay with appreciable rate because at least a quantum jump of two down in I₂ vibration is required for fragmentation in these cases. At this level of detail the

Table 3. Comparison of fluxes: A, calculated *via* the alternative RRKM theory with a zero-order separatrix surface; B, flux obtained by dynamical propagation of the zero-order separatrix; C, exact flux obtained by dynamical propagation of the exact separatrix

E/cm^{-1}	v	n	$F_s/2\pi\hbar^a$		
			A	B	C
107.3	1	3	0.38	0.25	0.19
235.3	2	3	0.53	0.37	0.25
363.3	3	3	0.68	0.46	0.32
491.3	4	3	0.81	0.51	0.37

^a The flux is divided by $2\pi\hbar$ to facilitate the discussion of the classical/quantum correspondence.

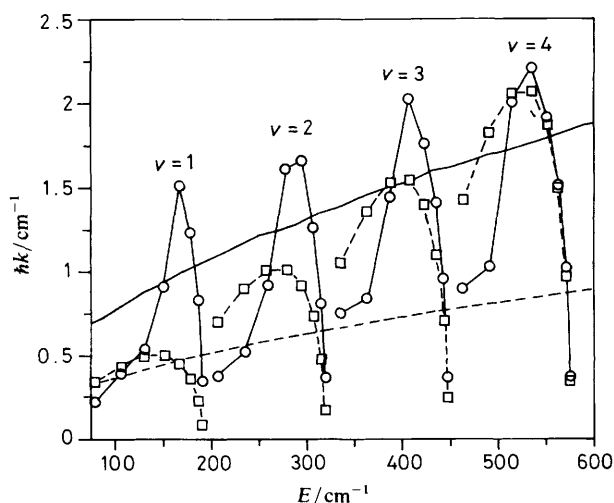


Fig. 3. Classical dynamical rate constants (—○—); Fermi golden rule rate constants (—□—); alternative RRKM rate constants (—); true intermolecular bottleneck or separatrix rate constants (---). See text for further discussion. The v manifolds associated with each inverted parabola pattern are labelled.

unimolecular fragmentation rate is very non-statistical, since it certainly does not increase monotonically with energy. Beswick and Jortner¹ noted, but did not discuss in detail, the behaviour just described. A simple explanation involves consideration of matrix elements of the form

$$\langle n | \exp[-2\alpha_{XB}(R - \bar{R})] | \epsilon' \rangle$$

which enter into the evaluation of eqn (35). The main contribution¹⁵ to such integrals tends to come from the region of the inner turning point of a zero-order classical trajectory evolving under V_R with translational energy ϵ' . Now, as n increases $\langle R | n \rangle$ becomes more peaked near the corresponding outer/inner turning points of a zero-order trajectory with energy $\epsilon_R(n)$. Assuming a steep potential, one expects more favourable overlap as n increases. However, as n becomes larger and closer to its limiting value, it is the nature of Morse oscillator eigenfunctions²⁰ that there is more amplitude in the outer turning point region than the inner turning point region and thus the magnitude of the above overlap integral will start to decrease.

The open circles in fig. 3 represent the results of the classical trajectory simulations. Remarkably, these results also show the same qualitative behaviour as do the quantum-mechanical calculations. It is useful to discuss these classical dynamical rate constants in relation to the classical statistical theory predictions. Fig. 3 displays, as a solid, monotonically increasing curve, the alternative RRKM theory rate constant. The dashed curve below the alternative RRKM result represents a scaling of the alternative RRKM curve by *ca.* 1/2 and, given the discussion in the first paragraph of this section, reflects the true intermolecular bottleneck statistical rate, devoid of any approximations to the transition state. In each *v* manifold the first few classical states, corresponding to *n* = 2, 3 and 4, do tend to lie near the intermolecular bottleneck result, reflecting the fact that the intermolecular bottleneck is the only relevant bottleneck.

Actually, for low *n* values one might expect intramolecular bottleneck effects to be relevant. Davis²¹ was the first to discuss the nature and construction of intramolecular bottlenecks for a problem of chemical interest. In the T-shaped HeI₂ system, intramolecular bottlenecks can exist within the intermolecular bottleneck and at low levels of excitation lead to appreciable deviations of the observed dynamical rate constants from statistical rate constants based on just the intermolecular bottleneck.^{2,3} Intramolecular bottlenecks are also partial barriers^{21,22} to the flow of classical phase points from one region of phase space to another and can be thought of as the remnants of certain tori, or quasibound motions. If the rate of flow of phase points out of an intramolecular bottleneck is much smaller than out of the intermolecular bottleneck, and if many initial phase points lie within the intramolecular bottleneck, then the observed dynamical rate constant will reflect the influence of the intramolecular bottleneck more than that of the intermolecular bottleneck. Intramolecular bottlenecks tend to be most restrictive near regions of quasiperiodic motion. Since there tend to be quasiperiodic regions near small *n* excitations, evidence for which can be seen in how trajectories in fig. 1(b) avoid the central small *n* region, it is in fact possible to see some intramolecular bottleneck effects when initial classical ensembles with *n* = 1 are examined. For example, more 'trapping' on surface of section plots can be observed, and lifetime distributions (lifetime distribution = $-\dot{P}_s$) exhibit apparent non-statistical behaviour biased toward longer lifetimes. The *n* = 1 rate constants are not displayed in fig. 3 because the energies of (*n* = 1, *v*) states lie in the region of the (*n* = 9, *v* - 1) states and therefore lead to a more complicated picture. The classical dynamical rate constants for *n* = 1 ensembles are small, ranging from $\hbar k = 0.07$ to 0.6 cm^{-1} as *v* ranges from 1 to 4; the corresponding quantum rate constants, as noted above, are in fact negligible. The classical rate constants just noted for *n* = 1 are all well within a factor of two of the intermolecular bottleneck rate constants of fig. 3, so intramolecular bottleneck effects are still not as severe as in some of the HeI₂ excitations studied previously.^{2,3}

Fig. 3 shows that the classical rate constants for moderate *n* values, such as *n* = 5, 6 and 7, tend to be larger than the exact intermolecular bottleneck rate constant and often larger than the more approximate statistical theory rate constant. This behaviour is attributable to the presence of a direct component to the fragmentation dynamics. Davis and Gray³ noted that in the T-shaped HeI₂ system, at very high initial excitations in the I₂ degree of freedom a large portion of the initial trajectory ensemble dissociated after the complex underwent just one vibration, which is a violation of the hypothesis of rapid intramolecular energy exchange within the transition-state surface. The fundamental dynamical origin of such a non-statistical effect was discussed by Davis and Gray and is related to the initial trajectory ensemble intersecting certain large inner homoclinic oscillations [fingers within the separatrix of fig. 2(a)] that lead ultimately to a very direct dissociation. As the initial *n* value is increased the importance of the direct component becomes greater because the initial ensembles intersect thicker inner homoclinic oscillations. Thus, whereas for *v* = 1 and *n* = 3, say, only 3% of the trajectories dissociate after one complex oscillation, by *n* = 6 about 43% of the trajectories dissociate after one complex oscillation.

For classical ensembles with $n = 8$ and 9 , fig. 3 shows that the classical rate constant starts to decrease. The origin of this behaviour is interesting. The quasiclassical trajectory ensembles naturally weight the inner and outer turning point regions more than other regions, with the outer turning point region being more heavily weighted than the inner turning point region for larger n . Moreover, for ensembles with the larger n values, the outer turning point region is so far away from the region of significant coupling that the actual trajectories, until they escape this region, evolve approximately along the corresponding zero-order trajectories. However, these trajectories then spend a long time in the vicinity of the outer turning point, which is the generic behaviour of zero-order trajectories. Trajectories near the inner turning point do not suffer this fate because the dynamical coupling is much larger there. The above considerations lead to a large proportion of the ensemble trajectories, for large n , taking a relatively longer time to fragment, leading to the observed dip in the classical rate constant. It is also important to note that the direct component discussed in the above paragraph is still responsible for most of the dissociation. For example, over 69% of the trajectories for $v = 1$, $n = 9$ dissociate *via* the direct component. Thus the existence of a direct component is not synonymous with a large rate of decay.

Further insight into the non-statistical behaviour in the classical rate constant as n increases can be obtained from inspection of some of the $\ln P_s(t)$ curves. Fig. 4 shows results for $v = 2$, with n being fixed at 3, 6 and 9. The $n = 3$ result is quite linear, consistent with the fragmentation being dominated by a single decay rate. The $n = 6$ result in fig. 4 is not as linear as $n = 3$. Careful inspection of fig. 4 suggests that there are possibly two decay modes: the first decay mode dominates up to $t \approx 2$ ps and is faster (steeper slope) than the second decay mode that exists for times > 2 ps. The fitted rate reflects both these modes. The initial decay here arises from the direct component discussed in the above two paragraphs. The $n = 9$ case of fig. 4 is also dominated by the direct component, although the qualitative behaviour of $\ln P_s(t)$ is very different, owing to the outer turning point effect discussed above. The reader will note, of course, that the assumption of a simple exponential decay rate in the $n = 9$ case is rather suspect. However, a simple exponential decay is a reasonable fit to the fragmentation data between *ca.* 2 and 10 ps. For example, we have also calculated rates using a 63% survival probability criterion for the maximum time sampled in the linear fits, which allows for a larger time range to be sampled. The classical dynamical rate constants for $n = 9$ typically change by only 10 to 15%.

It appears, from the above classical trajectory and Fermi golden rule results, that the rate of fragmentation of ArI_2 complexes is highly non-statistical. As one progresses up the n manifold serious violations of the rapid intramolecular energy exchange assumption occur. Consider, however, an ensemble of phase points that fills evenly the volume within the separatrix transition-state surface. By definition, the initial decay of these points out of the separatrix is exactly the alternative RRKM rate, whether or not rapid distribution of phase points has occurred. Thus, one might expect that the rate constants averaged over the n manifold

$$\langle k \rangle_v = \frac{1}{n_{\text{MAX}}} \sum_n k_{vn} \quad (36)$$

would correlate with the alternative RRKM rate constant. Of course eqn (36) is an average over an energy width and therefore does not correspond exactly to the ensemble average that would agree perfectly with the alternative RRKM rate constant. However, for small manifold energy width, as is the case for van der Waals molecules, one still expects a reasonable correlation. Table 4 compares the alternative RRKM rate constant, evaluated at an energy midway up each n manifold, to the corresponding classical and quantum-averaged manifold results. Thus, (a) the statistical theory rate constant agrees reasonably well with the averaged classical and quantum rate constants and also (b) the classical and quantum rate constants tend to agree as v is increased. However, it

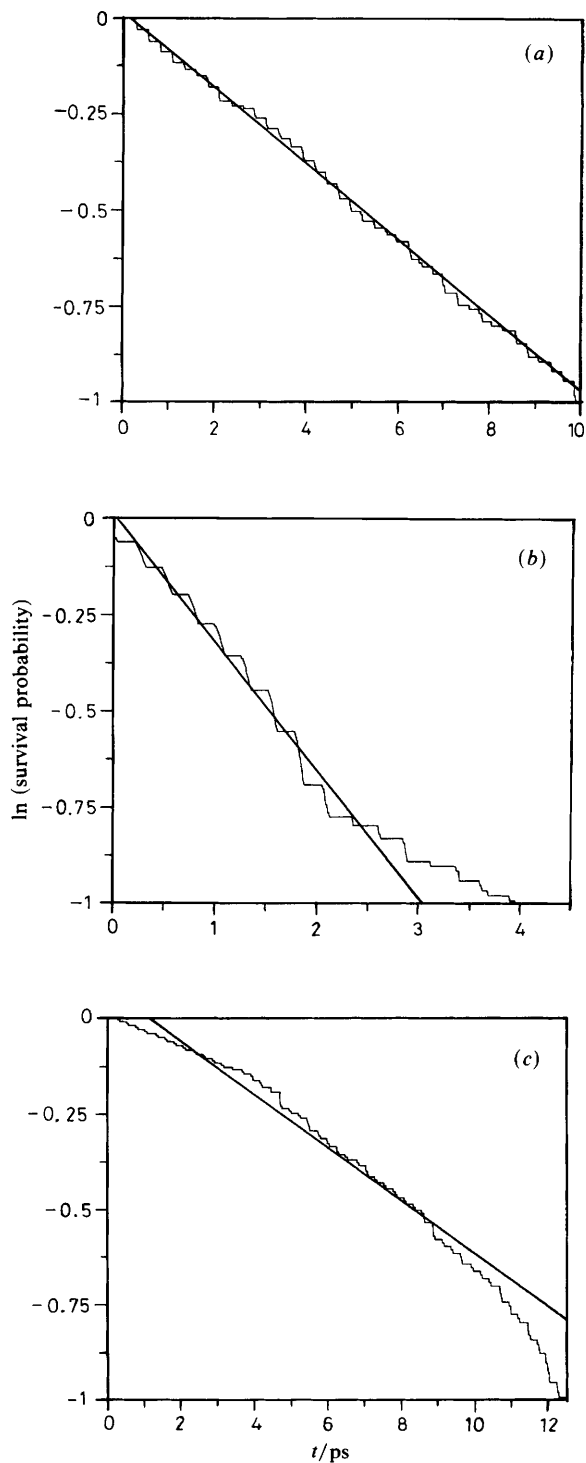


Fig. 4. Natural logarithm of the survival probability, $\ln p_s(t)$, for quasiclassical initial states $(v, n) = (a) (2, 3)$, $(b) (2, 6)$ and $(c) (2, 9)$. The corresponding total energies are 235.3, 295.3 and 319.5 cm^{-1} .

Table 4. Comparison of n -manifold averaged classical (CL), Fermi golden rule, (FGR) and alternative RRKM theory (ARRKM) rate constants

v	$\hbar k/\text{cm}^{-1}$		
	CL	FGR	ARRKM
1	0.67	0.32	0.76
2	0.87	0.66	1.17
3	1.13	0.99	1.38
4	1.29	1.38	1.64

is important to realize that the statistical theory rate constant, as noted at the beginning of this section, is only *ca.* a factor of two different from the rate constant calculated from the flux across the exact separatrix and so the agreement in table 4 is somewhat fortuitous. We conclude, therefore, that to within *ca.* a factor of two the average dynamical rate constant agrees with the statistical theory rate constant.

The agreement between manifold averaged dynamical rate constants and the statistical theory rate constants in table 4 can be thought of as another example of how statistical theories can provide useful information even when the fundamental assumption (iii) of the Introduction is not always satisfied. In essence, the average dynamical rate constants calculated above may be thought of as being the rate constants associated with initial phase point ensembles covering roughly evenly, *i.e.* statistically, the interior of the separatrix transition-state surface. (Of course, this is not precisely the case here because an energy width is being considered and also because the quasiclassical ensembles weight inner and outer turning point regions more than other regions.) These observations are also related to the work of Waite and Miller,²³ who argued on semiclassical grounds that standard RRKM theory predicts certain manifold averaged dynamical rate constants.

One way to understand why the classical statistical theory rate constants should agree with appropriately averaged quantum decay rate constants is to consider the magnitude of the flux F_s that enters into the statistical rate calculation. The factor $F_s/2\pi\hbar$ may be equated^{13,21} to the 'number of states at the transition state' in the semiclassical limit $\hbar \rightarrow 0$, so one could argue that for eqn (2) to be a reasonable description of the quantum dynamics, $F_s/2\pi\hbar$ must be such that at least one bound state exists, *i.e.* $F_s/2\pi\hbar$ must be at least 0.5 or more. The data in table 3 suggest that this is not yet the case for the energies examined in ArI₂, with the exact flux ranging from *ca.* a factor of three to a factor of 1.3 less than 0.5. However, this magnitude for the flux is larger than for most of the corresponding T-shaped HeI₂ fluxes,^{2,3} suggesting that the ArI₂ system may behave more classically than HeI₂. Indeed, the classical/quantum correspondence observed in table 4 for ArI₂ is better than that for the T-shaped HeI₂ system studied earlier,¹⁴ as well as that for a three-dimensional HeI₂ studied by Delgado-Barrio *et al.*,²⁴ where the classical dynamical rate constants were, at best, a factor of three larger than the quantum rate constants. When $F_s/2\pi\hbar \ll 0.5$, dynamical tunnelling may dominate the quantum rate constant.^{3,14,21} There is no obvious potential barrier in dynamical tunnelling. Nonetheless, it is interesting to draw parallels with Miller's²⁵ potential-barrier tunnelling correction to RRKM theory. For example, in Miller's theory the number of states at the transition state can become less than unity, in which case it is replaced by a tunnelling probability.

The above explanation for classical/quantum correspondence in van der Waals fragmentation is very qualitative and far from complete. For example, a curious result in the literature is that of Woodruff and Thompson,²⁶ who examined fragmentation of

a collinear HeI_2 model. The corresponding fluxes are undoubtedly similar to those for fragmentation of T-shaped HeI_2 . Yet these authors found classical rate constants in much better accord with quantum rate constants than we do.

Concluding Remarks

The dynamics associated with the fragmentation of van der Waals complexes have been discussed with the aid of exact classical trajectories, approximate quantum mechanical Fermi golden rule and statistical theory calculations of the rate constant. In particular, it has been shown that: (i) the statistical theory rate constant, which is based on using approximate phase space bottlenecks as transition states, can be reduced to a physically appealing form that reveals trends in the fragmentation rate; (ii) a non-monotonic variation in both the classical dynamical and approximate quantum-mechanical rate constants occurs for a collinear model of ArI_2 fragmentation; (iii) despite non-statistical effects that lead to (ii) above, rate constants averaged over the van der Waals manifold of states agree reasonably well with the statistical theory rate constants; (iv) the classical/quantum correspondence for the ArI_2 model is reasonably good and improves as the effective I_2 vibrational energy is increased.

In the near future we will report application of the theory described to other unimolecular reactions, in particular isomerization. We believe it to be worthwhile to explore variational criteria^{8,22,27} for obtaining better approximate separatrices. We also plan to explore quantum analogues of the phase-space bottlenecks, which can be studied in a variety of ways, including for example, projection operator methods, such as used in the formalism of Nordholm and Rice.²⁸

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